

Exercise 1.

Find the probability density function of $Y = e^X$ when X is normally distributed with parameters μ and σ^2 . Hint: pdf of the normal distribution: $f(x) = 1/\sqrt{2\pi\sigma^2} \exp[-(x - \mu)^2/(2\sigma^2)]$

Exercise 2.

Batteries of type A normally distributed with mean voltage 6.0 (volts) and variance 0.0225 (volts²). Type B batteries also normally distributed with mean voltage 12.0 and variance 0.04. If we form a series connection containing one of each type what is the probability that the combined voltage exceeds 17.4?

Exercise 3.

Mitchell and Alex are competing together in a 2-mile relay race. The time Mitchell takes to finish (in hours) is $X \sim \text{Exp}(2)$ and the time Alex takes to finish his mile (in hours) is continuous $Y \sim \text{Unif}(0, 1)$. Alex starts immediately after Mitchell finishes his mile, and their performances are independent. What is the distribution of $Z = X + Y$, the total time they take to finish the race?

Exercise 4.

Let X_1 and X_2 be independent Poisson random variables with each X_i having parameter λ_i . Find (a) $P(X_1 X_2 = 0)$; (b) $P(X_1 + X_2 = 1)$; (c) $P(X_1 + X_2 > 1)$.

Exercise 5.

Let X have a standard normal distribution. What is the pdf of $Y = X^2$? If you are going to use the formula, check if conditions are met.

Exercise 6.

Let $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$ be two independent exponential random variables. Find the probability density function of the random variable $Z = X + Y$. (Hint: pdf of $\text{Exp}(\lambda)$: $\lambda e^{-\lambda x}$, $x > 0, \lambda > 0$)